INDEPENDENT DATA ANALYTICS PORTFOLIO

Education Data Quality & Analytics

An End-to-End Python, SQL and Power BI Case Study

Project domain	Education monitoring, evaluation and learning
Programming	Python (Pandas) for ETL and Data Validation
Database	SQLite and advanced SQL for aggregation
Visualization	Interactive Power BI Dashboards
Processing result	180 records assessed; 175 accepted; 5 rejected (Synthetic Data)
Repository	**github.com/YOUR_USERNAME/YOUR_REPO**

Executive Summary

In the development sector, raw data from the field is rarely ready for policy decisions. Monitoring and Evaluation (MEL) systems frequently face invalid dates, missing consent forms, and anomalous enrolment figures. If these anomalies bypass quality checks, the resulting dashboards can mislead Ministry officials and donors, leading to misallocated resources.

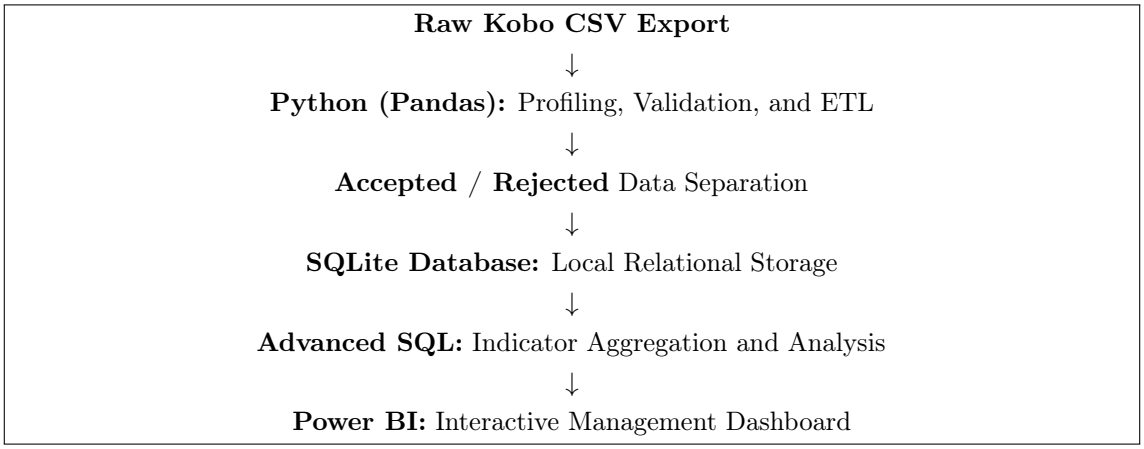
Note: The dataset used in this project is entirely synthetic, generated to simulate a real-world MEL environment while protecting privacy.

This project demonstrates the complete lifecycle of a Data Analyst: transforming 180 raw school-monitoring submissions into a secure, relational SQLite database, extracting actionable insights via advanced SQL, and delivering these insights through an interactive Power BI dashboard.

The Python pipeline rejected 5 records (2.8%) for critical data errors. The SQL analysis revealed that **seven out of eight districts are falling below the 85% attendance monitoring threshold**. By calculating the exact percentage-point gap from the target, the pipeline categorizes districts into "Priority Follow-up" and "Further Monitoring" tiers, providing management with a precise triage list for field interventions.

1. Technical Architecture

The workflow is designed to be reproducible and transparent, moving from raw field collection to executive visualization.



2. Data Quality: Safeguarding Policy Decisions

A functional MEL system cannot simply delete bad data; it must document and isolate it. By separating accepted records from rejected records, analysts ensure that subsequent policy decisions are based strictly on validated data.

The Python pipeline identified exactly five records for rejection. The table below shows the actual disposition of these synthetic test cases:

Record Anomaly	Validation Rule	Analytical Interpretation for Managers
Invalid date format	Valid Calendar Date	The reported date was 2026-13-45. This is a structural error. The record must be returned to the enumerator for correction before inclusion in time-series analysis.
Negative enrolment	Non-negative Integer	Boys' enrolment was reported as −50. This is mathematically impossible and corrupts district averages.
No consent given	Mandatory Policy	The consent field was "no". This is not a data entry error; it is a policy exclusion. The record must be securely archived and excluded from all reporting.
Malformed ID	Standardized Format	Identifier "SCH-47" fails the `RW-2026-####` pattern. This breaks traceability.
Duplicate Record	Unique ID	The record ID was submitted twice, likely due to a sync error.

Table 1: Validation rules and their operational implications for the 5 rejected records.

The MEL Principle: A quality flag identifies a condition to address; it does not always establish the cause. For instance, the district value "gasabo" (lowercase) was corrected via normalization in Python and *accepted*, rather than rejected.

3. Relational Database Management with SQLite

While Python is the ideal tool for cleaning and transforming data, it is insufficient for an institutional policy environment. Policy-makers require a data structure that provides speed, consistency, and traceability.

I loaded the 175 clean records into a **SQLite** relational database. While SQLite does not provide enterprise-level access controls, in this local analytical environment, it serves three critical functions:

1. **Immediate Responsiveness:** When a Director asks, *"Which districts are failing the attendance target?"*, SQL allows the analyst to filter and aggregate records in milliseconds.
2. **Traceability:** By retaining the original source CSV, the validation log, and versioned SQL scripts, we create a documented audit trail. If a donor questions a figure, the exact query and underlying records can be traced.
3. **BI Integration:** The database acts as the single source of truth, feeding data into visualization tools.

3.1 Advanced SQL for MEL Indicators

In MEL, how you write the SQL query directly impacts the validity of the indicator. A common mistake is using `AVG(attendance_rate)`, which calculates an *unweighted* average. This means a school with 10 students carries the same statistical weight as a school with 1,000 students.

To calculate a true, weighted district-level indicator and categorize the management response, the SQL query must sum the numerators and denominators first, and use conditional logic:

```
SELECT
    district,
    COUNT(*) AS total_visits,
    SUM(enrol_total) AS total_enrolment,
    SUM(attend_total) AS total_attendance,
    ROUND(100.0 * SUM(attend_total) / NULLIF(SUM(enrol_total), 0), 1)
        AS attendance_pct,
    ROUND((100.0 * SUM(attend_total) / NULLIF(SUM(enrol_total), 0)) - 85, 1)
        AS gap_from_85_pct_points,
    CASE
        WHEN SUM(enrol_total) = 0 THEN 'Check denominator'
        WHEN 100.0 * SUM(attend_total) / SUM(enrol_total) >= 85 THEN 'On track'
        ELSE 'Below target'
    END AS management_status
FROM clean_submissions
GROUP BY district
ORDER BY attendance_pct DESC;
```

4. Visualizing the Data: Power BI for Management

Python and SQL speak the language of analysts. Managers, however, need visual stories to make rapid decisions. I connected Power BI to the SQLite data to create an interactive dashboard.

4.1 Illustrative Management Use Case

(Note: The following is an illustrative scenario of how the dashboard is intended to be used, not a real quotation.)

Imagine a programme manager reviewing the dashboard. Instead of reading a static spreadsheet, they click on a specific district slicer (e.g., Muhanga). The dashboard instantly updates to show its attendance rate and enrolment ratios. The visual red flags allow the manager to see immediately which districts require attention, shifting the conversation from *"what are the numbers?"* to *"how do we allocate our field monitors?"*.

4.2 Technical Integration

Because Power BI uses an import model, the dashboard reflects the state of the database at the time of import. Updating the visuals requires a manual or scheduled data refresh. This distinction is important when reporting to stakeholders, as it clarifies that the dashboard is a snapshot, not a live-query feed.

5. Comprehensive Policy Findings

The SQL database was used to extract the complete performance landscape across all eight districts. To provide management with actionable triage, the data was analyzed using the 85% operational threshold.

5.1 Attendance Performance and Gap Analysis

The SQL query calculates the exact percentage-point gap from the 85% target. This allows managers to distinguish between districts that are close to the threshold and those in critical need of intervention.

District	Visits	Attendance	Gap from 85%	Management Status
Kicukiro	22	87.4%	+2.4	On Track
Huye	17	84.2%	-0.8	Below Target
Rubavu	29	81.7%	-3.3	Below Target
Gasabo	29	78.3%	-6.7	Priority Follow-up
Nyagatare	18	75.9%	-9.1	Priority Follow-up
Nyarugenge	24	73.6%	-11.4	Priority Follow-up
Musanze	16	72.9%	-12.1	Priority Follow-up
Muhanga	20	67.1%	-17.9	Priority Follow-up

Table 2: Complete Attendance Landscape with Gap Analysis.

Strategic Insights:

- **Only Kicukiro (87.4%)** is currently meeting the 85% operational threshold.
- **Huye (84.2%) and Rubavu (81.7%)** are below target but require only *Further Monitoring*, as their gaps are relatively small.
- **Muhanga, Musanze, Nyarugenge, Nyagatare, and Gasabo** are flagged for *Priority Follow-up*. Muhanga is severely off-track at 67.1%—a massive 17.9 percentage points below the target.

Recommendation: The M&E team should immediately deploy field monitors to the five Priority Follow-up districts to investigate the root causes—whether it is a data collection error, environmental factors, or systemic absenteeism.

5.2 Enrolment Ratios and Equity Analysis

To measure equity (SDG 4.5.1), the SQL query calculated the ratio of enrolled girls to enrolled boys. A perfect parity is 1.00.

District	Girls	Boys	Ratio	Observation
Nyagatare	6,405	4,699	**1.36**	Higher proportion of girls
Rubavu	9,277	7,100	**1.31**	Higher proportion of girls
Muhanga	5,831	4,767	1.22	Higher proportion of girls
Kicukiro	5,885	5,143	1.14	Slight female advantage
Musanze	4,873	4,298	1.13	Slight female advantage
Huye	4,541	4,153	1.09	Near parity
Gasabo	8,637	8,002	1.08	Near parity
Nyarugenge	7,455	7,468	**1.00**	Perfect parity

Table 3: Girls-to-Boys Enrolment Ratios.

Analytical Interpretation and Follow-up Hypothesis:

The enrolment data shows a higher number of enrolled girls than boys in several districts. Nyagatare has a girls-to-boys enrolment ratio of approximately 1.36, while Rubavu has a ratio of approximately 1.31.

These ratios identify differences in recorded enrolment; they do not, on their own, establish dropout rates or explain the underlying causes. Differences may reflect population composition, the education levels covered, reporting completeness, or other contextual factors.

Recommended follow-up: Validate the enrolment data, disaggregate results by education level and age where available, and investigate attendance and longitudinal retention records. Consult school leaders and communities before selecting targeted interventions, as these disparities may indicate male dropout driven by agricultural labour, migration, or other unobserved factors.

6. Limitations and Reproducibility

To maintain professional integrity, the following limitations must be acknowledged:

- **Synthetic Data:** The anomalies (e.g., 2026-13-45) were artificially injected into the test dataset to demonstrate the validation pipeline.
- **Descriptive Scope:** This analysis is strictly descriptive. It shows *what* is happening, not *why*. Causal inference requires a baseline, a control group, and qualitative research.
- **Proxy Indicators:** The ratio calculated here is based on raw enrolment. The official SDG 4.5.1 indicator requires rigorous alignment with UNESCO methodologies for specific education levels.
- **Threshold Definition:** The 85% attendance threshold is used here as an operational triage tool. It is not an official Ministry of Education mandate.

How to Reproduce:

1. Run `mel_pipeline.ipynb` in Jupyter Notebook to generate the synthetic raw data and clean it.
2. Open `education_mel.db` in DB Browser for SQLite to view the relational data.
3. Open `dashboard.pbix` in Power BI Desktop to view the interactive visuals (requires a data refresh to pull local data).

7. Conclusion

This project demonstrates that an effective Data Analyst does far more than clean data. They design robust validation rules, architect databases that ensure institutional memory, write advanced SQL queries that expose hidden patterns, and deliver those insights through clear, interactive Power BI dashboards.

By transforming 180 raw submissions into a structured SQLite database and using SQL to calculate the exact gap from the attendance target, this pipeline provides the technical evidence required to prioritize field interventions and explore new hypotheses regarding educational equity in rural Rwanda.

Reliable data. Transparent indicators. Better-informed decisions.